

Effect of Market Volatility on ARIMA Forecasting Accuracy for Pension NAV Time Series

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Abstract

This micro research note tests whether market volatility degrades ARIMA forecasting accuracy for a pension NAV time series. Volatility is proxied using the rolling standard deviation of NAV returns, and two equal-length windows ($N=60$ each) are selected: a low-volatility “Normal” window (2023-04-26 to 2023-07-21) and a high-volatility “Volatile” window (2024-04-02 to 2024-07-03). ARIMA(p,d,q) orders are selected per window via AIC, and out-of-sample forecasts are evaluated using MAE, RMSE, and MAPE over a 12-step horizon. Forecast error increases during volatile conditions (MAE: $0.1076 \rightarrow 0.2004$; RMSE: $0.1463 \rightarrow 0.2492$; MAPE: $0.98\% \rightarrow 1.41\%$). The results indicate ARIMA’s predictive performance weakens under volatility regime shifts, motivating volatility-aware or hybrid approaches for NAV planning.

Keywords: ARIMA, forecasting accuracy, volatility regime, pension NAV, time series

1. Problem

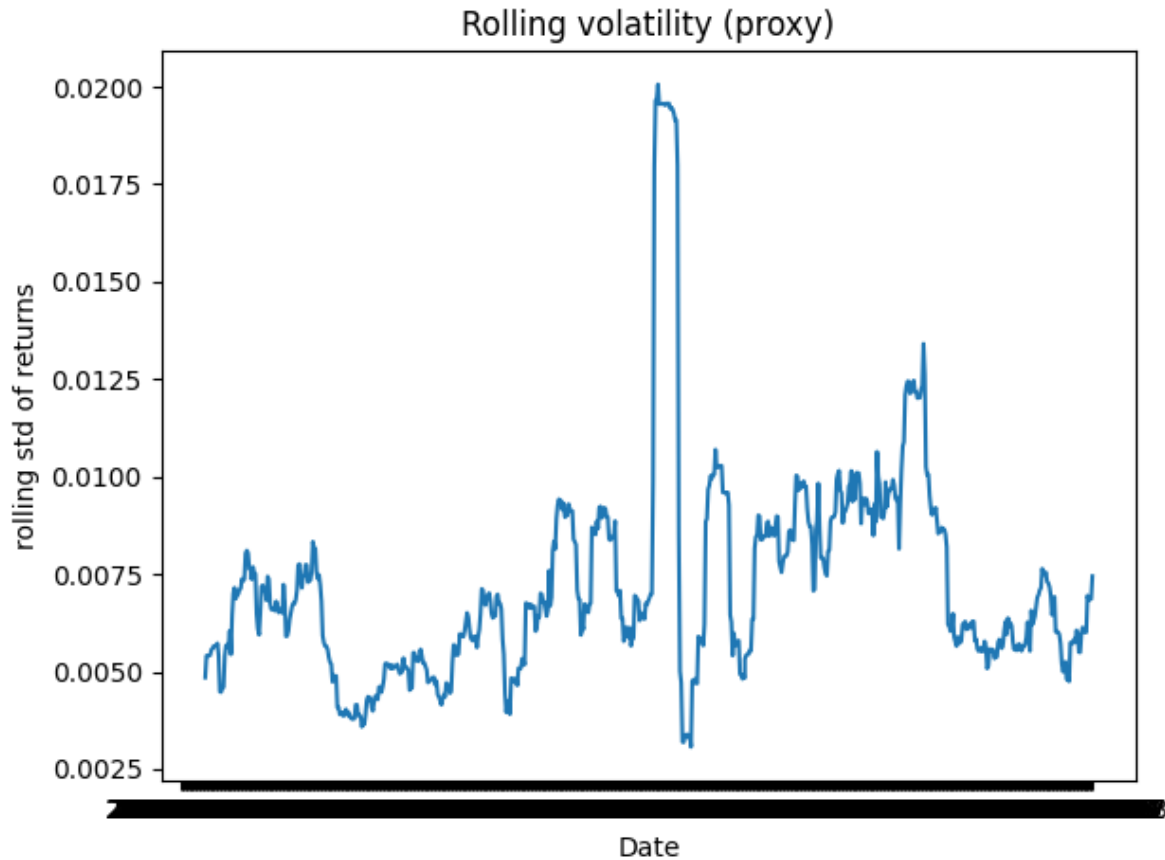
Pension fund Net Asset Value (NAV) time series often exhibit regime shifts where returns become more volatile during stress periods. ARIMA models are widely used for short-horizon forecasting, but their performance may degrade when the underlying volatility structure changes rapidly. This note asks: how does ARIMA forecasting error change during high-volatility NAV windows?

2. Data and Volatility Proxy

Fund/portfolio: Axis Pension Fund. Frequency: Daily. Full period covered: 2022-10-25 to 2026-01-28.
Data source: Downloaded from the Axis Pension Fund website.

Returns are computed as percentage change in NAV. Volatility is proxied using the rolling standard deviation of returns with window length 20.

Figure 1. Rolling volatility (proxy) computed as rolling standard deviation of returns.



3. Method

Two windows of equal length ($N=60$ observations each) are selected from the volatility series: a low-volatility Normal regime and a high-volatility Volatile regime. Within each window, ARIMA(p,d,q) order is selected by minimizing AIC over a candidate grid (implemented in the accompanying notebook). Each window is split into train/test by holding out the last 12 observations (forecast horizon). Models are fit on the train segment and forecast 12 steps ahead; errors are computed on the held-out test segment using MAE, RMSE, and MAPE (%).

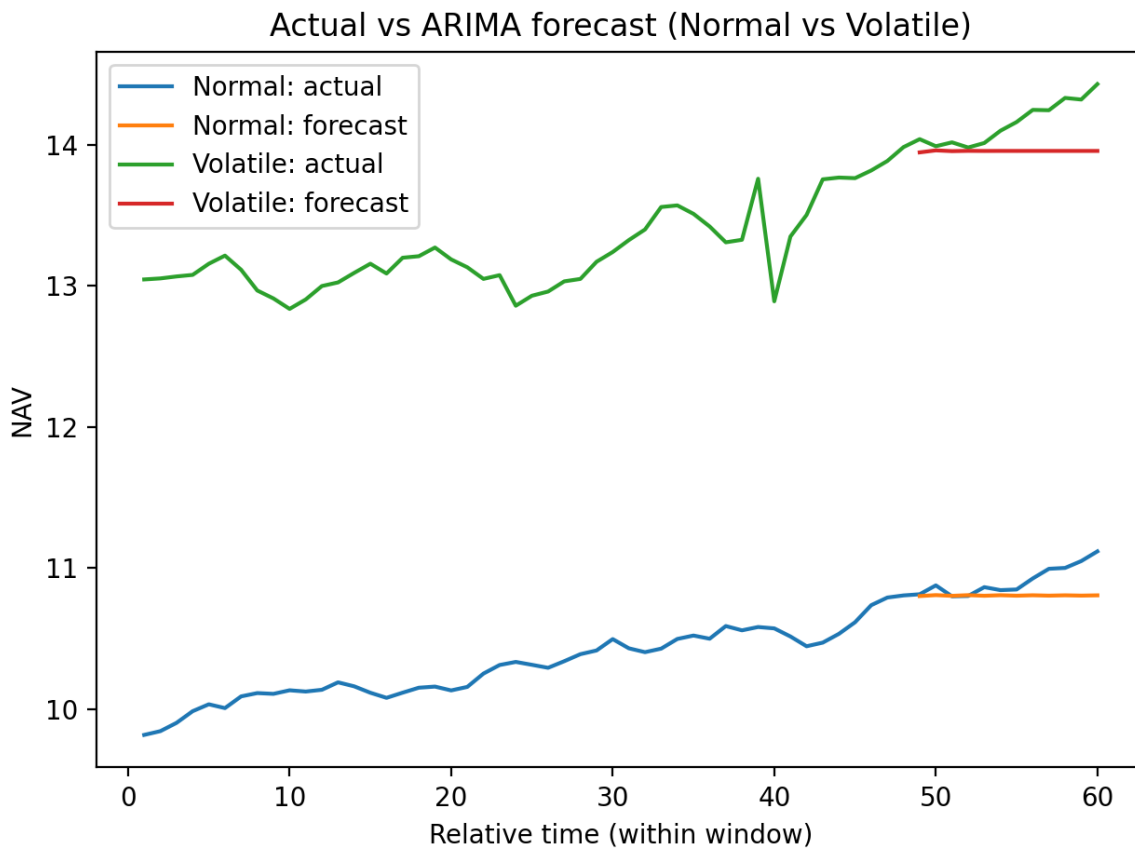
4. Results

Table 1. Forecast accuracy of ARIMA across regimes ($N=60$ each). Orders selected by AIC within each window.

Regime	Window	N	ARIMA(p,d,q)	AIC	MAE	RMSE	MAPE (%)
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Normal	2023-04-26 to 2023-07-21	60	(2, 1, 2)	-189.5587	0.1076	0.1463	0.9777
Volatile	2024-04-02 to 2024-07-03	60	(1, 1, 0)	-51.5883	0.2004	0.2492	1.4051

Figure 2. Actual vs ARIMA forecast using relative time within each window (removes calendar gap between regimes). Forecasts are shown for the final 12-step hold-out horizon.



In the Normal regime, ARIMA(2,1,2) achieves MAE=0.1076, RMSE=0.1463, and MAPE=0.9777%. In the Volatile regime, ARIMA(1,1,0) yields higher error with MAE=0.2004, RMSE=0.2492, and MAPE=1.4051%. Relative to the Normal window, volatility increases error by $\sim 1.86\times$ (MAE), $\sim 1.70\times$ (RMSE), and $\sim 1.44\times$ (MAPE).

5. Interpretation

Forecast error rises materially in the high-volatility window, consistent with ARIMA's sensitivity to abrupt changes in local trend and variance. Since ARIMA extrapolates from recent autocorrelation patterns, volatility regime shifts can reduce the stability of the model's residual dynamics, producing larger forecast deviations. The effect is consistent across MAE, RMSE, and MAPE, suggesting it is not metric-specific.

Limitation: ARIMA orders are selected separately per regime; results reflect best-fit ARIMA per regime rather than a single fixed specification.

6. Conclusion

ARIMA forecasting accuracy decreases during high-volatility NAV regimes. For practical NAV planning, volatility-aware methods (e.g., regime switching, ARIMAX with market drivers, or hybrid models) may offer improved robustness during stress periods.

7. Reproducibility and Availability

Code and notebooks: <https://github.com/ajaysaxenabtech/micro-research-note-arima-volatility>

Data availability: NAV data was downloaded from the Axis Pension Fund website; users should verify redistribution permissions for the raw dataset.

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